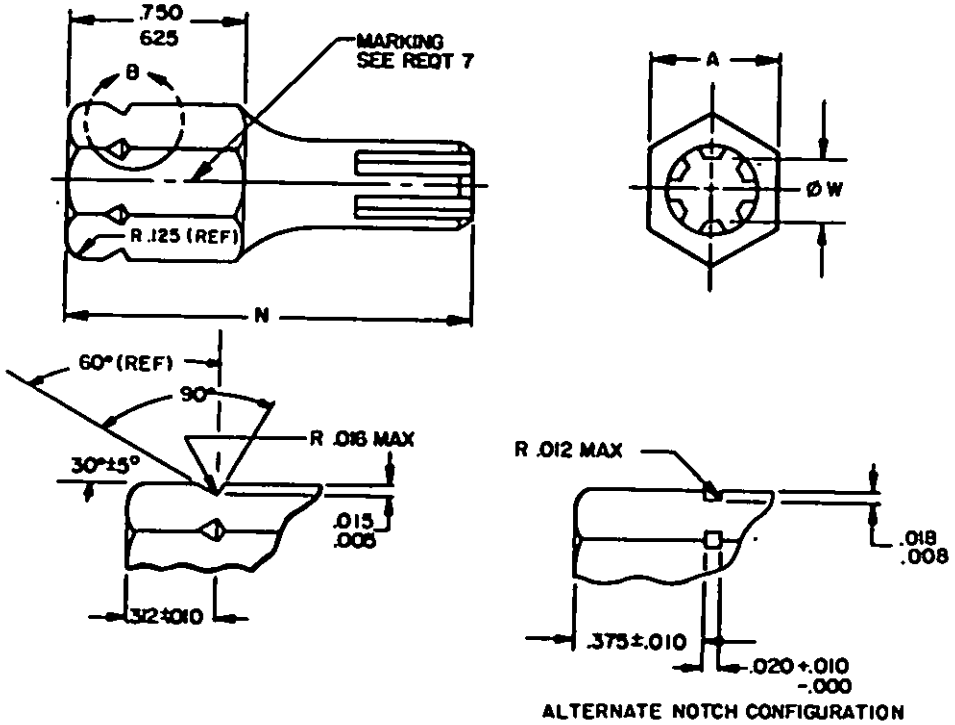
MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, SIX SPLINE, STRAIGHT, EXTERNAL HEX SHANK, MANUALLY DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the bits described herein shall consist of this specification and the latest issue of MIL-B-9946.



VIEW B-RETAINING GROOVE-NOTCH TYP SIX PLACES

NOTES:
1. All dimensions are in inches
2. Bit, retaining groove and tip point shall be in accordance with TABLE I.

FIGURE 1. BIT, SCREWDRIVER, SIX SPLINE, STRAIGHT, EXTERNAL HEX SHANK, MANUALLY DRIVEN

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	BIT DIMENSIONS								TIP DIMENSIONS								TORSIONAL MOMENT LBF - IN
	DRIVER SIZE MAX	RECESS SIZE NOM	SCREW DIAMETER SIZE NOMINAL REF				A HEX +.000 -.004	N LENGTH ±.120	Ø M MAJOR		Ø W MINOR		S WIDTH OF SPACE		L LENGTH STRAIGHT SURFACE	K CHAMFER	
			CAP SCREW	FLAT CTSK HEAD	BUTTON HEAD	SET SCREW			MAX	MIN	MAX	MIN	MAX	MIN	MIN	MAX	
096	.096	.096	#2 & #3	#5 & #6	#5 & #6	#8	.250	1.000	.0960	.0950	.0790	.0775	.0240	.0230	.125	.008	19
111	.111	.111	#4 & #5	#8	#8	#10	.250	1.000	.1110	.1100	.0940	.0925	.0280	.0270	.125	.009	29
133	.133	.133	#6	-	-	-	.250	1.000	.1330	.1310	.1140	.1120	.0340	.0320	.156	.014	50
145	.145	.145	-	#10	#10	1/4	.250	1.000	.1450	.1435	.1240	.1225	.0355	.0340	.187	.015	65
168	.168	.168	#8	-	-	-	.250	1.000	.1680	.1660	.1440	.1420	.0410	.0390	.216	.016	111
183	.183	.183	#10	1/4	1/4	5/16	.250	1.000	.1830	.1815	.1580	.1565	.0440	.0425	.216	.016	131
216	.216	.216	1/4	5/16	5/16	3/8	.312	1.312	.2160	.2145	.1840	.1825	.0550	.0535	.250	.022	216
251	.251	.251	-	3/8	3/8	7/16	.312	1.312	.2510	.2495	.2140	.2125	.0655	.0640	.281	.024	329
291	.291	.291	5/16	7/16	7/16	1/2	.312	1.312	.2910	.2895	.2460	.2445	.0775	.0760	.312	.030	530

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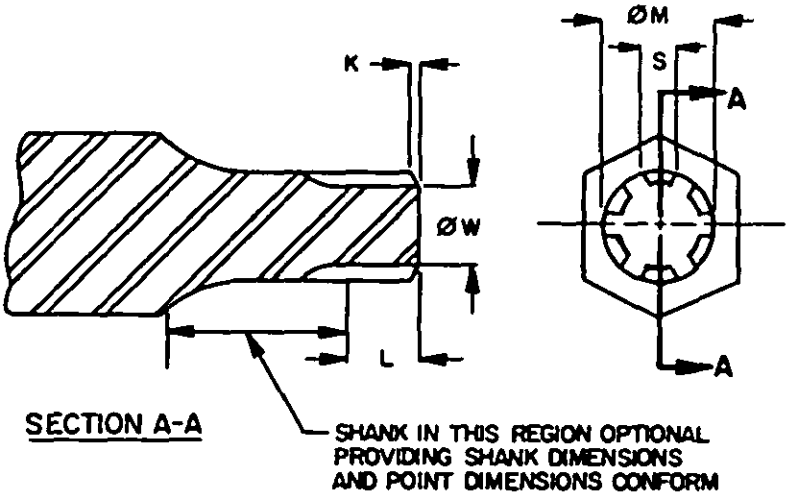


FIGURE 2. TIP POINT

REQUIREMENTS:

1. Material: Alloy Tool Steel in accordance with American Society for Testing Material (ASTM) A681, Shock-Resisting Steel, Type S-2, (UNS T41902).

2. Hardness: Macrohardness, (Rockwell C Scale) 56-62HRC. Three macrohardness readings shall be taken at the location shown in FIGURE 3. Each hardness value must be within the range specified.

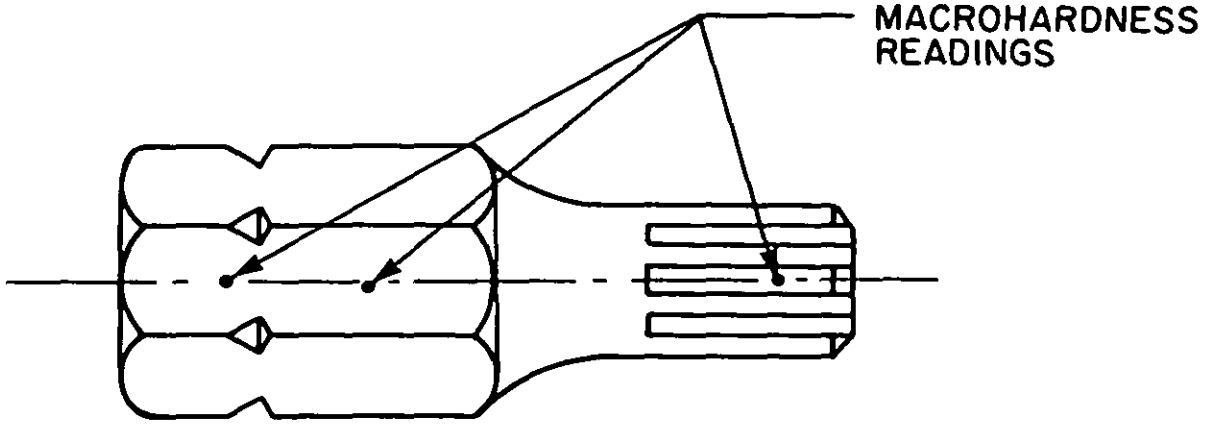


FIGURE 3. MACROHARDNESS READING LOCATIONS

Superficial hardness (Rockwell HR15N scale) 87-90. Superficial hardness readings shall be taken on bid samples at two locations on sectionalized specimens along the blade.

3. Driver bit gage: Driver bit gage shall be in accordance with FIGURE 4 and TABLE II.

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4. Tip point: Tip point shall be in accordance with FIGURE 2 and TABLE I.

5. Test block: Test block shall be in accordance with FIGURE 5 and TABLE III. Torque tests are conducted using this block as described herein and in MIL-B-9946.

6. Torsion test: When tested, all bits shall withstand the test load specified in TABLE I without injury or permanent deformation which may affect their durability or serviceability. A torsion test shall be conducted to determine conformance with the test load specified in TABLE I. The bit shall be inserted into a female socket designed for the size bit being tested. The bit shall withstand the torsional moment (test load) specified in TABLE I. An angular reference line shall be established and the bits shall show less than a 5° permament angular deflection. The bit shall show no flaws, cracks or other defects which might affect its durability or serviceability. The part of the bit inserted into the socket shall show no distortion or rounded corners. For bit sizes .096 inches thru .251 inches, inclusive, the torsional load shall be further increased until the bit fails. Failure shall occur by a uniform shear fracture for these sizes. Sample bits shall be tested using a test recess conforming to FIGURE 5 and TABLE III. There shall be no debris present in the recess during the test.

7. Identification marking: Driver bits shall be marked by etching, stamping or engraving onto each bit the following minimum information:

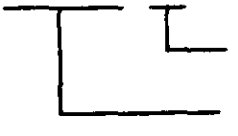
   a. Military specification part number.

   b. Manufacturer's symbol or code.

   c. Country of manufacture.

8. Part number: The part number consists of the letter "M", the basic number of this specification sheet, followed by a dash number taken from TABLE I.

Example: M9946/10-183



M9946/10-183 indicates - Bit, Screwdriver, Six Spline, Straight, External Hex Shank, Manually Driven; Driver Size Max .183; Recess Size Nom .183; Screw Diameter Size Nom Ref, #10, 1/4, and 5/16; Hex Size .250; Bit Length 1.000; Tip Major Diameter .1830 Max - .1815 Min; Minor Diameter .1580 Max - .1565 Min; Length .216; Chamfer .016.

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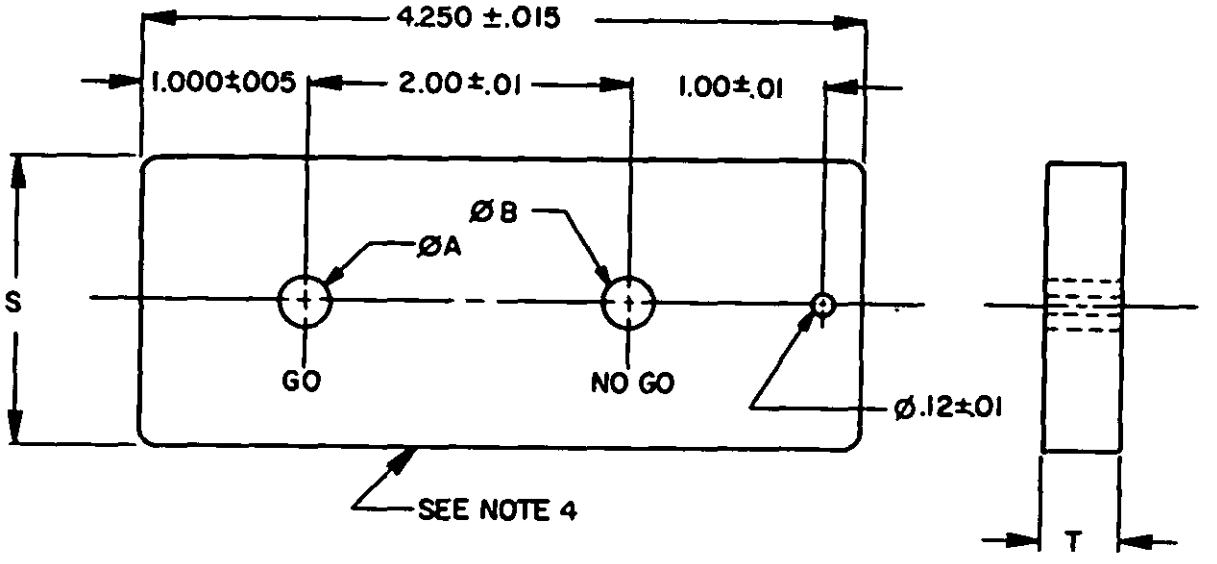


FIGURE 4. DRIVER BIT GAGE

TABLE II. DRIVER BIT GAGE DIMENSIONS FOR SIX SPLINE BITS

KEY AND SOCKET SIZE NOMINAL	GO ØA MAX	GO ØA MIN	NO GO ØB MAX	NO GO ØB MIN	S MIN	T ±.005
.096	.0963	.0960	.0950	.0947	1.00	.187
.111	.1113	.1110	.1100	.1097	1.00	.187
.133	.1333	.1330	.1310	.1307	1.00	.187
.145	.1453	.1450	.1435	.1432	1.00	.187
.168	.1683	.1680	.1660	.1657	1.00	.375
.183	.1833	.1830	.1815	.1812	1.00	.375
.216	.2163	.2160	.2145	.2142	1.00	.375
.251	.2513	.2510	.2495	.2492	1.00	.375
.291	.2913	.2910	.2895	.2892	1.00	.625

NOTES:
1. All dimensions are in inches.
2. Material: CRES in accordance with ASTM A276, Type 440C (UNS S44004).
3. Hardness: 58-62HRC
4. Marking: Stamp nominal key and socket size and "GO" and "NO GO".

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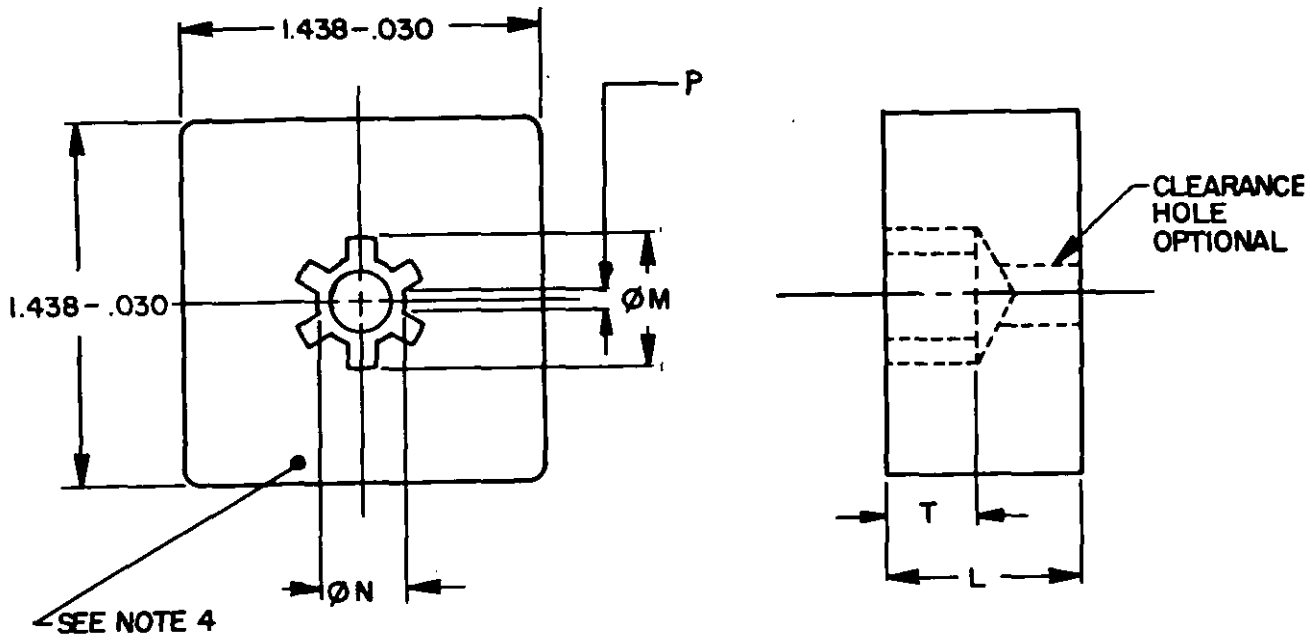


FIGURE 5. TEST BLOCK

TABLE III. TEST BLOCK DIMENSIONS FOR SIX SPLINE BITS

KEY AND SOCKET SIZE NOMINAL	Ø M MAJOR		Ø N MINOR		P WIDTH OF SPACE		T MINIMUM ENGAGEMENT STRAIGHT SIDES	L THICKNESS REF
	MAX	MIN	MAX	MIN	MAX	MIN		
.096	.098	.097	.082	.080	.022	.021	.096	.187
.111	.115	.113	.098	.096	.025	.023	.111	.187
.133	.137	.135	.118	.116	.030	.028	.133	.187
.145	.149	.147	.128	.126	.032	.030	.145	.187
.168	.173	.171	.150	.147	.036	.033	.168	.375
.183	.188	.186	.163	.161	.039	.037	.183	.375
.216	.221	.219	.190	.188	.050	.048	.216	.375
.251	.256	.254	.221	.219	.060	.058	.251	.375
.291	.298	.296	.254	.252	.068	.066	.291	.750

NOTES:
1. All dimensions are in inches.
2. Material: Alloy Steel in accordance with ASTM A681, Type S-2 (UNS T41902).
3. Hardness: 62-64HRC
4. Marking: Stamp nominal key and socket size.

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